

NATIONAL INSTITUTE OF ENGINEERING

END SEMESTER EXAMINATION - MAY 2027

Duration: 3 Hours

Max Marks: 80

Instructions: Attempt any four questions. Figures to the right indicate full marks.

Q1(a) Explain the layered reference model and justify each layer with a diagram. [6]

Q1(b) Describe the scheduling policy and derive its average waiting time. [6]

Q1(c) Differentiate between the two normalisation forms with a worked example. [6]

Q2(a) Explain the transaction recovery procedure and describe how the

system restores a consistent checkpoint after an unexpected failure. [8]

Q2(b) Derive the expression for throughput and state all assumptions clearly. [6]

Q2(c) Compare the two indexing structures and evaluate their lookup complexity. [6]